EXP 2

Create labeled matrices (5×4, 3×3, 2×2) filled by row/column.

AIM:

To create labeled matrices (5×4, 3×3, 2×2) filled by row/column.

PROCEDURE:

Step 1: Open RStudio or R console on your system.
Step 2: Create a matrix by arranging a sequence of numbers into rows and columns. Step 3: Display the created matrix to check its elements and layout.
Step 4: Create another matrix with a different size (for example, 3×3) to understand matrix creation with various dimensions.
Step 5: Display the second matrix to verify its arrangement.
Step 6: Create a smaller matrix (for example, 2×2) to observe how matrix size can vary. Step 7: Display the third matrix to view its content.
Step 8: Observe that each matrix organizes elements in a rectangular form with equal-length rows and columns.

CODE:

```r
matrix1 <- matrix(1:20, nrow = 5, ncol = 4, byrow = TRUE)
rownames(matrix1) <- c("R1", "R2", "R3", "R4", "R5")
colnames(matrix1) <- c("C1", "C2", "C3", "C4")
cat("5x4 Matrix filled by row:\n")
print(matrix1)
matrix2 <- matrix(1:9, nrow = 3, ncol = 3, byrow = FALSE)
rownames(matrix2) <- c("R1", "R2", "R3")
colnames(matrix2) <- c("C1", "C2", "C3")
cat("\n3x3 Matrix filled by column:\n")
print(matrix2)
matrix3 <- matrix(1:4, nrow = 2, ncol = 2, byrow = TRUE)
rownames(matrix3) <- c("R1", "R2")
colnames(matrix3) <- c("C1", "C2")
cat("\n2x2 Matrix filled by row:\n")
print(matrix3)
```

OUPUT:

```
5x4 Matrix filled by row:
   C1 C2 C3 C4
R1  1  2  3  4
R2  5  6  7  8
R3  9 10 11 12
R4 13 14 15 16
R5 17 18 19 20

3x3 Matrix filled by column:
   C1 C2 C3
R1  1  4  7
R2  2  5  8
R3  3  6  9

2x2 Matrix filled by row:
   C1 C2
R1  1  2
R2  3  4
```